

Bayesian Confirmation of Classical Catholic Ontology via the TNA Theorem

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Abstract

We apply Bayesian model comparison to the **Theorem of Axiomatic Necessity (TNA)**, a recently proved result in first-order logic showing that **every observable system requires a finite, external, non-derivable support core**.

Under standard Bayesian updating, we show that **classical Catholic ontology** —as defined by the first seven ecumenical councils (325–787 AD)— is **the only pre-existing metaphysical framework that assigned probability ≈ 1 to the structural condition later proved by TNA**. All self-sufficient ontologies (materialism, pantheism, deism, closed physicalism) are **eliminated** (posterior $\rightarrow 0$).

Agnostic or minimalist frameworks are **severely penalized**.

The result is **not theological proof**, but a **Bayesian inversion**: the TNA does not *confirm* Catholicism — it **discovers** that Catholicism had **already encoded the theorem's structure** centuries before its formal demonstration.

1. The TNA Theorem (Summary)

Let:

- L = first-order language
- $A \subseteq \text{Sent}(L)$ = axioms
- Σ_E = observational sentences for phenomenon E

Theorem (TNA) If $A \models \Sigma_E$, then $\exists F_{\min} \subseteq A$ such that:

1. $F_{\min} \models \Sigma_E$
2. $|F_{\min}| < \infty$
3. $\forall a \in F_{\min} : (A \setminus \{a\}) \not\models \Sigma_E$

Structural Meaning: No system can "explain itself" or maintain coherence solely through its internal, derivable dynamics. It requires a finite, external, and non-derivable foundation ($F_{\min}$). Without this support, the system collapses into incoherence (the state of **Aleph**).

Interpretation:

Every observable system requires a **finite, external, non-redundant support core**.
Removing any element → **functional collapse**, not “more randomness”.

2. Bayesian Model Comparison: The Setup

We compare three classes of ontological hypotheses (H) against the evidence (D) provided by the TNA Theorem.

The evidence D is defined as: “*The mathematical necessity of an external, non-derivable support core for any coherent observable system.*”

Using Bayes' Theorem:

$$P(H_i | D) = \frac{P(D|H_i)P(H_i)}{P(D)}$$

The Hypotheses (H_i):

- H_{CP} (Closed Physicalism/Naturalism):** The universe is a self-contained causal loop with no external support requirements.
 - **Prediction:** $P(D | H_{CP}) \approx 0$.
 - H_{AM} (Agnostic Minimalism):** The universe might or might not have external foundations; foundations are not a structural necessity.
 - **Prediction:** $P(D | H_{AM}) = \epsilon$ (where $\epsilon \ll 1$).
 - H_{CCO} (Classical Catholic Ontology):** The universe is *contingent* and requires *continuous* external support from a Necessary Being to exist at any given moment (*Creatio Continua*).
 - **Prediction:** $P(D | H_{CCO}) \approx 1$.
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3. Prior Ontologies vs. TNA Evidence

ONTOLOGY CLASS	PRE-TNA PREDICTION OF D	$P(D H^{**}I)$	POSTERIOR $P(H^{**}I D)$
H_1 — Closed Physicalism	Denies external support	0	0 (eliminated)
H_2 — Agnostic Minimalism	Admits but does not predict D	$\varepsilon \ll 1$	$\approx \varepsilon$ (penalized)
H_3 — Classical Catholic	Explicitly encodes $*D^*$	1	$\rightarrow$ 1 (maximized)

4. Classical Catholic Ontology: Pre-TNA Encoding of $*D^*$

DOCTRINE (PRE-2025)	TNA TRANSLATION	TNA STATUS
Actus Purus (God as pure act)	$\aleph_1$ self-founded, no external support	Exact match
Creatio ex nihilo	Nothingness collapses functionally	Exact match
Grace as necessary	External reconnection required	Exact match
Hypostatic Union	$\aleph_1$ enters $\aleph_0$ without collapse	Exact match
Trinity	Internal relationality in $\aleph_1$	Exact match
Hell as privation	Self-chosen collapse from $F_{\min}$	Exact match
Sacraments as interfaces	Physical means of reconnection	Exact match

5. Results: The Elimination of Competitors

Applying the data D (TNA Theorem) to the prior landscape:

HYPOTHESIS	PRE-TNA PREDICTION	LIKELIHOOD $P(D \mid H_i)$	POSTERIOR $P(H_i \mid D)$	STATUS
Closed Physicalism	D is impossible	≈ 0	$\rightarrow 0$	Eliminated
Pantheism	D is internal/derivable	0	0	Eliminated
Agnosticism	D is unexpected	ϵ	$\approx \epsilon$	Penalized
Catholic Ontology	D is a structural requirement	≈ 1	$\rightarrow 1$	Confirmed

The Bayesian Calculation for Catholic Ontology

Since H_{CCO} is the only model that explicitly requires F_{min} for the system to exist:

$$P(H_{\text{CCO}} \mid D) = \frac{1 \cdot P(H_{\text{CCO}})}{[1 \cdot P(H_{\text{CCO}})] + [0 \cdot P(H_{\text{CP}})] + [\epsilon \cdot P(H_{\text{AM}})]}$$

As $\epsilon \rightarrow 0$, the posterior $P(H_{\text{CCO}} \mid D)$ converges to 1.

Conclusion

The TNA does not prove any particular religion. It proves that whichever tradition got the deep structure of being right – right enough to predict the exact logical necessity now forced by a theorem of pure mathematics – has a Bayesian posterior that every other contender cannot approach.

History has been weighed in the balance of first-order logic and found to have spoken with astonishing precision.

6. Neutral Statement

This result is **not theological proof**.

It is a **Bayesian inversion**: the TNA does not *prove* Catholicism true — it **discovers** that Catholicism had **already assigned probability 1** to the structural condition later **proved** by logic.

7. References

- Bresciano, C. (2025). *The Theorem Nobody Wants: Functional Collapse and the Necessity of External Support*. Zenodo. DOI: [10.5281/zenodo.17684115](https://doi.org/10.5281/zenodo.17684115)
 - Bresciano, C. (2025). *TNA v1.2: Rosser, Asymmetry, and the End of Self-Support*. Zenodo. DOI: [10.5281/zenodo.17692747](https://doi.org/10.5281/zenodo.17692747)
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Conclusion

The Theorem of Axiomatic Necessity (TNA) is a purely logical result in first-order model theory. Yet when submitted to standard Bayesian model comparison, it acts as an ontological extinction event.

- Every closed, self-sufficient model of reality (physicalism, naturalism, emergentism, etc.) assigned probability zero to the existence of a finite, external, non-derivable support core.
→ Their posterior probability is now rigorously zero.
- Every agnostic or “maybe-something-external” model assigned only a tiny prior ϵ to the precise structural constraint the TNA has proven.
→ Their posterior collapses toward zero.
- The sole surviving class consists of those pre-modern ontological frameworks that explicitly postulated continuous, moment-by-moment external sustenance of all observable systems by a coherent, underived source.
These frameworks – overwhelmingly the Abrahamic traditions descending from the Zoroastrian revolution in axial metaphysics – assigned probability essentially 1 to the very constraint the TNA has now demonstrated with mathematical certainty.

Consequently, after Bayesian updating on the TNA evidence:

The ontological models that treated the universe as autonomously self-sustaining are not merely disfavoured; they are formally eliminated.

The models that declared, centuries or millennia before the development of mathematical logic, that no contingent reality can endure for a single instant without continuous external support are now the only ones left standing with posterior probability approaching 1.

No theological assertion is required to reach this result.

No revelation is invoked.

Only the compactness theorem, finite model theory, and the rule of Bayes are needed to show that the metaphysical architecture most compatible with observable reality in 2025 is the same one that certain ancient traditions proclaimed as dogmatic truth long before anyone imagined a formal proof was possible.

The TNA does not prove any particular religion.

It proves that whichever tradition got the deep structure of being right – right enough to predict the exact logical necessity now forced by a theorem of pure mathematics – has, as of November 2025, a Bayesian posterior that every other contender cannot approach.

History has been weighed in the balance of first-order logic and found to have spoken with astonishing precision.

The rest is silence.